3.30 Fractional Ideals

Definition 3.30.1
Let A be an integral domain contained in its quotient field K. An A-submodule $I \subset K$ is called a ***fractional ideal*** *if there exists $a \in A$ such that $aI \subset A$.*

We say a fractional ideal is ***principal*** *if it is generated by some $x \in K$ as an A-module.*

Proposition 3.30.2 (Operations on Fractional ideals)
Let A be an integral domain and $I, J \subset K$ fractional ideals. Then the following constructions are also fractional ideals

a) $I + J = \{i + j \mid i \in I j \in J\}$
b) $IJ = \left\{\sum_{k=1}^{n} i_k j_k \;\middle|\; i_k \in I j_k \in J\right\}$
c) $(I : J) = \{x \in K \mid xJ \subseteq I\}$

Proof. It is straightforward to prove that these constructions are again A-submodules of K.

Suppose that $aI \subseteq A$ and $bJ \subseteq A$, then $(ab)(I + J) \subseteq A$ and $(ab)IJ \subseteq A$. Similarly

$$(I : J)J \subseteq I \implies j(I : J) \subseteq I \implies abj(I : J) \subseteq A$$

for some $0 \neq j \in J$ ∎

Proposition 3.30.3
Let A be an integral domain. The collection of fractional ideals $\mathcal{I}_A$ forms a monoid under multiplication of ideals, with identity A.

Definition 3.30.4
Let A be an integral domain and I a fractional ideal. Then we say I is ***invertible*** *if there exists a fractional ideal J such that $IJ = A$.*

Proposition 3.30.5
Let A be an integral domain and I a fractional ideal. Then I is invertible if and only if $I(A : I) = A$, in which case $(A : I)$ is the unique inverse.

Proof. Uniqueness follows from general properties of monoids (...). If $I(A : I) = A$ then by definition I is invertible. Conversely suppose I is invertible, that is $IJ = A$. Then evidentally $J \subseteq (A : I)$ so $A = IJ \subseteq I(A : I) \subseteq A$. Therefore $(A : I)$ is an inverse for I as required. ∎

Corollary 3.30.6
Let A be an integral domain. Then the invertible fractional ideals form an abelian group under multiplication.

Definition 3.30.7 (Localisation of Fractional Ideals)
Let A be an integral domain contained in its quotient field K. Let I be a fractional ideal and S a multiplicatively closed subset of A. Then define

$$S^{-1}I := \left\{\frac{i}{s} \;\middle|\; i \in Is \in S\right\}$$

If $\mathfrak{a}$ is an ideal then $S^{-1}\mathfrak{a}$ corresponds to the usual notion (3.6.17) and is equal to the extension of $\mathfrak{a}$ under the inclusion homomorphism $A \to S^{-1}A$.

If $\mathfrak{p}$ is a prime ideal of A then as usual define

$$I_{\mathfrak{p}} := (A \setminus \mathfrak{p})^{-1} I$$

and we may write

$$\mathfrak{a}_{\mathfrak{p}} = \mathfrak{a} A_{\mathfrak{p}}$$

Proposition 3.30.8 (Localisation of Fractional Ideals)
Let A be an integral domain contained in its quotient field K. Let S be a multiplicatively closed subset of A and I a fractional ideal. Then regarding as subrings of K then $S^{-1}I$ is a fractional ideal of $S^{-1}A$. Furthermore

a) $S^{-1}(I + J) = S^{-1}I + S^{-1}J$
b) $S^{-1}(IJ) = (S^{-1}I)(S^{-1}J)$
c) $S^{-1}(I : J) = (S^{-1}I : S^{-1}J)$

Proof. Evidently $S^{-1}I$ is again a fractional ideal of $S^{-1}A$.

a) Clear because $\frac{i+j}{s} = \frac{i}{s} + \frac{j}{s}$.
b) We have $\frac{ij}{s} = \left(\frac{i}{s}\right)\left(\frac{j}{1}\right)$ whence $S^{-1}(IJ) \subseteq (S^{-1}I)(S^{-1}J)$. Conversely we have $\left(\frac{i}{s}\right)\left(\frac{j}{t}\right) = \frac{ij}{st}$ so we ahve the reverse inclusion.
c) Suppose that $\frac{x}{s} \in S^{-1}(I : J)$ and $\frac{j}{t} \in S^{-1}J$. Then $\left(\frac{x}{s}\right)\left(\frac{j}{t}\right) = \frac{xj}{st} \in S^{-1}I$ because $x \in (I : J)$. Therefore $S^{-1}(I : J) \subseteq (S^{-1}I : S^{-1}J)$. Conversely given $x \in (S^{-1}I : S^{-1}J)$ and $j \in J$. Then by definition $x\left(\frac{j}{1}\right) = \frac{i}{s}$ whence $sx \in (I : J)$. Therefore $x \in S^{-1}(I : J)$ as required.

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Lemma 3.30.9
Let A be an integral domain contained in its quotient field K. Suppose I is a fractional ideal then

$$I = \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$$

In particular we have

$$A = \bigcap_{\mathfrak{m}} A_{\mathfrak{m}}$$

Proof. Evidently $I \subseteq \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Suppose that $x \in \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Consider the ideal $\mathfrak{a} := \{a \in A \mid ax \in I\}$. Suppose $\mathfrak{a}$ is proper then by (3.3.15) it is contained in a maximal ideal $\mathfrak{m}$. As $x \in I_{\mathfrak{m}}$ we have $(A \setminus \mathfrak{m}) \cap \mathfrak{a} \neq \varnothing$ and $\mathfrak{a} \subseteq \mathfrak{m}$ a contradiction. Therefore $\mathfrak{a} = A$ and in particular $x = 1.x \in I$ as required. ■

Proposition 3.30.10 (Invertibility is a local property)
Let A be an integral domain with quotient field K, and I a fractional ideal of A. Then the following are equivalent

a) I is invertible
b) $I_{\mathfrak{p}}$ is invertible for all prime ideals $\mathfrak{p}$
c) $I_{\mathfrak{m}}$ is invertible for all prime ideals $\mathfrak{m}$

Proof. a) $\implies$ b) By assumption and (3.30.5) $I(A : I) = A$ so by (3.30.8) $I_{\mathfrak{p}}\left(A_{\mathfrak{p}} : I_{\mathfrak{p}}\right) = A_{\mathfrak{p}}$, as required.

b) $\implies$ c) Every maximal ideal is prime (3.3.59) so we are done.

c) $\implies$ a) Let I be a fractional ideal then

$$\begin{aligned}
I(A : I) &= \bigcap_{\mathfrak{m}} (I(A : I))_{\mathfrak{m}} && (3.30.9)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A : I)_{\mathfrak{m}} && (3.30.8)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A_{\mathfrak{m}} : I_{\mathfrak{m}}) && (3.30.8)\\
&= \bigcap_{\mathfrak{m}} A_{\mathfrak{m}} && (3.30.5)\\
&= A && (3.30.9)
\end{aligned}$$

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